

From Algorithm to Map: Implementation Case Studies for Three Adoption Pathways

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Abstract

Algorithmic redistricting has three realistic adoption pathways: through an independent commission, through ordinary statute, and through court-ordered remedial relief. Each pathway places different institutional actors in the role of map adopter, raises different procedural questions, and invites different legal challenges. This paper develops concrete case studies for each pathway using the **redist** algorithmic redistricting system. Case Study 1 examines the Arizona Independent Redistricting Commission (AIRC), which already employs politically balanced membership and community-of-interest hearings; we analyze how the commission could integrate algorithm output into its existing workflow while preserving its adjustment authority. Case Study 2 examines the California Citizens Redistricting Commission (CRC), whose statutory criteria include proportionality and county contiguity requirements that map directly onto **redist**'s VRASection and county-sticky edge weights. Case Study 3 examines North Carolina post-*Harper v. Hall*, where the state supreme court's authority to appoint a special master and order a remedial map creates conditions under which algorithm output could constitute the remedy absent a demonstrated constitutional defect. A practical workflow table identifies who runs which command, at which stage, under what oversight. We conclude with a candid assessment of the challenges—data preparation burden, public transparency requirements, and challenge processes—that any adopting jurisdiction will face.

Keywords: redistricting, independent commission, court remedy, special master, Arizona, California, North Carolina, implementation

1 Introduction

Algorithmic redistricting is not merely a theoretical alternative to political map-drawing. It is a system that can be operationalized—run by identified actors, subject to identified oversight procedures, and producing outputs that enter an identified legal process. Abstracting away from implementation, as much of the literature does, leaves open the critical question of whether algorithmic redistricting is politically and institutionally feasible.

This paper answers that question by developing concrete case studies for three adoption pathways.

The Commission Pathway (Case Study 1 — Arizona AIRC) is the path of least resistance where an independent redistricting commission already exists. The commission could treat the algorithm’s output as a starting plan that satisfies mathematical criteria, then apply its community-of-interest adjustment authority to refine boundaries before public adoption.

The Statute Pathway (Case Study 2 — California CRC) involves legislative or commission action in a state that has already adopted multi-criterion redistricting criteria. California’s criteria—population equality, VRA compliance, county contiguity, compactness, and proportionality—map closely onto the **redist** system’s weights and constraints. A California-style adoption would require demonstrating that each criterion is satisfied algorithmically, a task the paper shows is tractable.

The Court-Order Pathway (Case Study 3 — North Carolina post-*Harper*) is available where state courts have found constitutional violations in enacted maps. The court-order pathway places the algorithm at the center of the remedial process: a special master appointed by the court runs **redist**, the output becomes the remedial map, and the burden shifts to the challenger to demonstrate a constitutional defect in the output.

These three pathways are not mutually exclusive. A state might adopt statutory criteria that require algorithmic verification, or a court might order a commission to use an algorithm. But treating them as distinct ideal types allows us to identify the institutional requirements, oversight structures, and legal challenges specific to each.

The paper is organized as follows. Sections 2–4 develop the three case studies in detail. Section 5 provides a practical workflow table that identifies, at each stage of the redistricting process, who runs which command, when, under what oversight, and with what review opportunity. Section 6 addresses challenges common to all three pathways: data preparation, public transparency, and the challenge process. Section 7 concludes.

Throughout, we use the **redist** system [Della-Libera, 2026b] as the reference implementation. The commands cited in the workflow table are drawn from the **redist** CLI

reference [Della-Libera, 2026d]. The legal analysis draws on Della-Libera [2026a].

2 Case Study 1 — Arizona AIRC (Commission Pathway)

2.1 Background: The AIRC Structure

The Arizona Independent Redistricting Commission was established by Proposition 106, a voter initiative adopted in 2000, following decades of legislative gerrymandering. The AIRC consists of five members: two Democrats, two Republicans, and one independent chair selected by the other four members from a pool maintained by the state commission on appellate court appointments. The Commission drew the 2002, 2012, and 2022 congressional maps.

The U.S. Supreme Court upheld the AIRC’s constitutional authority in *Arizona State Legislature v. Arizona Independent Redistricting Commission* [U.S. Supreme Court, 2015], holding that the Elections Clause’s reference to “the Legislature” encompasses a lawmaking body created by the People through the initiative process. The AIRC is therefore a constitutionally valid redistricting body for federal congressional maps.

Arizona’s redistricting criteria, in statutory priority order, are: (1) population equality; (2) VRA compliance; (3) geographic compactness and contiguity; (4) preservation of communities of interest; and (5) partisan fairness. The algorithm addresses criteria 1 and 3 directly, and criterion 2 through its VRASection weight; criteria 4 and 5 are the domain of Commission judgment.

2.2 How the AIRC Would Integrate Algorithm Output

Under the Commission pathway, algorithmic redistricting functions as a *constraint-satisfying starting plan*. The workflow would proceed in four stages:

1. **Data preparation and algorithm run.** Commission staff (or a designated technical contractor) run `redist state -state AZ -year 2020 -version v1` to generate a population-balanced, compact map. The VRASection weight is applied to protect minority-opportunity districts in the Phoenix metropolitan area. The output is a GEOID-to-district assignment file and a map visual.
2. **Criteria-compliance verification.** Commission staff run `redist analyze -state AZ -year 2020 -version v1 -types compactness demographic` to generate the criteria-

compliance report. This report documents population balance, compactness scores, and minority-population percentages for each district.

3. **Public comment period.** The Commission publishes the algorithm output as a draft plan and receives public comment, including community-of-interest testimony. The Commission retains authority to adjust boundaries to address specific community-of-interest concerns identified in testimony, subject to the constraint that adjustments must not reduce population balance below $\pm 1\%$ or reduce compactness scores below a floor.
4. **Final adoption.** The Commission votes on the final plan. If the algorithm output is adopted without adjustment, the Commission’s decision record documents that criteria 1–3 are satisfied by the algorithm and that criteria 4–5 are satisfied by the absence of challenged community separations and by the near-symmetric partisan lean of competitive districts.

2.3 Community-of-Interest Adjustments

The most delicate question is how the Commission can exercise judgment about communities of interest without re-introducing political manipulation. Two guardrails are essential.

First, community-of-interest adjustments must be *geographically justified*: they must unite a geographically defined community (a county, a Native American tribal area, an established municipality) rather than unite partisan voters who happen to live in dispersed locations.

Second, adjustments must be documented with a *specific factual finding* that the original algorithm output separated the identified community. Arizona’s statute already requires this; algorithmic redistricting strengthens the requirement by providing a clean baseline against which adjustments are measured.

2.4 Legal Resilience of the Commission Pathway

The Commission pathway is legally robust for several reasons. The AIRC’s constitutional authority is established. Algorithmic redistricting reduces the decisional surface available to commissioners, reducing opportunities for the kind of intentional-partisanship finding that would be required to sustain a challenge. And the transparency of the algorithm’s criteria—population equality and compactness, publicly documented—makes it difficult to allege hidden partisan purpose in the starting plan.

3 Case Study 2 — California CRC (Statute Pathway)

3.1 Background: The California CRC and Its Criteria

California’s Citizens Redistricting Commission was established by Proposition 11 (2008), extended to congressional redistricting by Proposition 20 (2010). Unlike the AIRC’s five-member structure, the CRC has 14 members: five registered Democrats, five registered Republicans, and four members registered with neither major party.

California’s redistricting criteria, in statutory priority order, are: (1) population equality (one person, one vote); (2) VRA compliance; (3) geographic contiguity; (4) minimization of city and county splits; (5) geographic compactness; and (6) preservation of communities of interest. California does not include partisan fairness as a statutory criterion, distinguishing it from Arizona.

3.2 Algorithmic Satisfaction of California’s Criteria

The `redist` system’s weight parameters map onto California’s criteria in a natural way.

Criterion 1 (population equality) is enforced directly: the algorithm requires population balance within $\pm 0.5\%$ of the equal-population target, stricter than the legal requirement for congressional districts.

Criterion 2 (VRA compliance) is addressed through `VRASection`, which increases the edge weight on adjacency edges connecting tracts with high minority-population fractions, encouraging the algorithm to keep minority communities together. Della-Libera [2026c] demonstrates that `VRASection` produces more majority-minority districts than the compactness-only baseline.

Criterion 3 (geographic contiguity) is an invariant enforced by the algorithm’s adjacency-graph construction: only geographically contiguous tracts are considered adjacent, so every output district is contiguous by construction.

Criterion 4 (minimization of city and county splits) corresponds to the county-sticky edge weight in the `redist` system. Increasing the weight on county-boundary adjacency edges penalizes splits. The county-sticky weight can be calibrated to California’s specific county structure, with larger counties (Los Angeles, San Diego) receiving higher weights to reflect their larger legislative significance.

Criterion 5 (geographic compactness) is the algorithm’s primary objective through METIS edge-cut minimization. Polsby-Popper and Reock compactness scores for California algorithmic districts exceed the CRC’s enacted map on average.

Criterion 6 (preservation of communities of interest) is addressed through the

Commission’s adjustment authority, as described for the AIRC in Section 2.

3.3 The Proportionality Consideration

California’s criteria do not include partisan proportionality as a statutory criterion, but CRC commissioners have historically considered proportionality as a non-binding guideline when evaluating competing plans. Algorithmic maps for California tend to produce partisan outcomes close to the statewide Democratic vote share (approximately 63% in 2020 presidential results) because they optimize geometry rather than partisan outcomes.

The relevant question is whether the CRC would find an algorithmic starting plan adequate. Given that the algorithm satisfies criteria 1–5 quantifiably and reproducibly, and given that criterion 6 is reserved for Commission judgment, the CRC would be in the same position as the AIRC: using the algorithm output as a constrained starting plan and applying community-of-interest adjustments where evidence supports them.

3.4 VRASection and California’s Minority Communities

California has substantial Latino, Asian American, and Black populations that generate VRA obligations. The Los Angeles Basin contains multiple majority-Latino census tracts; the San Francisco Bay Area contains significant Asian American communities.

VRASection addresses these concentrations by applying enhanced edge weights that reflect minority population fractions. The specific threshold for VRA protection can be set to reflect the CVAP (Citizen Voting Age Population) percentages required by VRA analysis rather than the total population. Della-Libera [2026c] and ? develop the calibration methodology.

Under California’s criteria ordering, VRA compliance ranks second (after population equality), so the VRASection weight is applied before the county-sticky weight in the algorithm’s optimization. This ensures that minority-opportunity districts are preserved even where county preservation would require splitting them.

4 Case Study 3 — North Carolina Post-*Harper* (Court-Order Pathway)

4.1 Background: *Harper v. Hall*

Harper v. Hall [Supreme Court of North Carolina, 2022] is the most significant state court partisan gerrymandering decision since *Rucho v. Common Cause* [U.S. Supreme Court, 2019]

closed the federal courthouse door. The North Carolina Supreme Court held, 4-3, that the North Carolina General Assembly’s enacted congressional and state legislative maps violated the state constitution’s free elections clause and equal protection guarantee. The court ordered the legislature to redraw the maps and retained jurisdiction to review the remedial plan.

The decision was subsequently subject to procedural complexity—a change in the court’s composition led to reconsideration—but *Harper* established the template for state-court-ordered redistricting remedies in the partisan gerrymandering context. This case study uses the *Harper* fact pattern as the model for the court-order pathway without taking a position on North Carolina’s subsequent procedural history.

4.2 The Special Master Role

When a court orders a redistricting remedy, it typically has three options: (a) order the legislature to redraw the map under specified criteria; (b) draw the map itself; or (c) appoint a special master to draw the map. Option (a) risks delay and noncompliance. Option (b) is constitutionally problematic as courts are not representative bodies. Option (c) has become the standard remedy in federal redistricting cases and is available to state courts as well.

The algorithmic redistricting system is particularly well-suited to the special master context. A special master appointed by the court could be instructed to:

1. Run `redist state -state NC -year 2020 -version remedy` with population-balance and compactness objectives.
2. Apply VRASection weights to protect the Black-majority district in the northeastern part of the state (the historical Gingles area) and any other minority-opportunity district required by VRA analysis.
3. Submit the algorithm output to the court as the proposed remedial plan, with a criteria-compliance report generated by `redist analyze -state NC -year 2020 -version remedy -types compactness demographic political`.
4. Defend the plan in court proceedings, addressing any constitutional challenge to the output.

4.3 The Algorithm Output as the Presumptive Remedy

The court-order pathway suggests a strong procedural innovation: the algorithm output should be presumed constitutional absent a specific showing of constitutional defect. This

presumption reflects two structural features of algorithmic redistricting.

First, the algorithm cannot gerrymander because it cannot see partisan data. A constitutional challenge on partisan-gerrymandering grounds would need to show that the *output* constitutes a constitutional violation regardless of the process—a much harder showing than demonstrating partisan intent.

Second, the algorithm satisfies the neutral criteria (population equality, compactness) that every court has endorsed as constitutional. A challenger must therefore show not merely that a different map would be more favorable but that the algorithm’s output violates a specific constitutional requirement.

This structure shifts the burden of proof in a legally significant way. Rather than the state defending a legislatively drawn map against allegations of partisan manipulation, the special master defends an algorithmically drawn map against allegations of specific constitutional defects. The latter is a substantially easier task.

4.4 Scope of Judicial Review

The court-order pathway does not eliminate judicial review; it changes its focus. Judicial review of an algorithmic remedial map would appropriately address:

- Whether the algorithm’s population-balance constraint is adequate (the equal-population requirement).
- Whether the VRASection weighting adequately protects minority-opportunity districts (VRA compliance).
- Whether the algorithm’s compactness objective is consistent with state constitutional criteria (state law compliance).
- Whether any community-of-interest exceptions identified by the special master are supported by factual findings.

Questions of partisan outcome—which party benefits, what the expected seat count is—would be legally irrelevant under *Rucho* in the federal context and, after *Harper*, would need to be evaluated against the specific constitutional standard adopted by the state court. Crucially, because the algorithm does not access partisan data, no partisan intent can be attributed to the mapmaker; partisan-outcome challenges must rest on effects rather than intent.

5 Practical Workflow

Table 1 presents a practical workflow for all three adoption pathways, identifying the responsible actor, the `redist` command, and the oversight mechanism at each stage.

Table 1: Practical Workflow: Three Adoption Pathways

Stage	Commission (AZ/CA)	Statute	Court Order (NC)
Data prep	Commission staff; <code>redist fetch</code>	State GIS office; <code>redist fetch</code>	Special master staff; <code>redist fetch</code>
Initial run	Staff run <code>redist state</code> ; Commission re-views output	Designated agency runs <code>redist state</code> ; Legislative committee reviews	Special master runs <code>redist state</code> ; Court receives report
Analysis	<code>redist analyze</code> compactness + demographic	<code>redist analyze</code> all types	<code>redist analyze</code> political + demographic
Public comment	60-day public hearing period; Commission adjustments logged	Legislative hearing; Public comment period	14-day objection period; Parties file briefs
Adjustment	Commission vote on adjustments; documented against algorithm baseline	Agency or commission adjusts; legislative approval	Special master proposes adjustments; court approves
Final adoption	Commission vote	Legislative vote or commission order	Court order
Review	State court (if challenged)	State court (if challenged)	Higher court (appeal)

Note: `redist fetch` downloads and validates census data. `redist state` runs the partitioning algorithm. `redist analyze` generates criteria-compliance reports. Commands follow the `redist` CLI reference [Della-Libera, 2026d].

5.1 Key Differences Across Pathways

The three pathways differ primarily in three dimensions.

Who runs the algorithm. In the commission pathway, commission staff or a designated technical contractor runs the algorithm under the commission’s supervision. In the statute pathway, a designated state agency (e.g., the state GIS office or a redistricting board) runs the algorithm. In the court-order pathway, the special master or the master’s technical support team runs the algorithm. In all three cases, the algorithm is run openly, with full documentation of inputs, parameters, and outputs.

How adjustments are made. Commission pathways allow the commission to adjust the algorithm output based on community-of-interest testimony. Statute pathways may delegate adjustment authority to an agency or reserve it to the legislature. Court-order pathways confine adjustments to those supported by specific constitutional findings, and the special master’s adjustment proposals are subject to judicial review.

What counts as a successful challenge. In the commission and statute pathways, successful challenges require showing that the commission or agency exceeded its authority or violated a criterion. In the court-order pathway, the challenge must show a constitutional defect in the output—a higher bar that makes the court-order pathway the most legally stable of the three.

6 Challenges and Solutions

All three adoption pathways face common implementation challenges. This section addresses the three most significant: data preparation, public transparency, and the challenge process.

6.1 Data Preparation

The algorithm requires census tract boundary files and population counts, both available from the U.S. Census Bureau’s TIGER/Line and redistricting data programs. Preparing these inputs for a production run involves:

1. **Downloading census data.** The `redist fetch -state AZ -year 2020` command automates downloads from the Census Bureau. For larger states (Texas, California, New York), this may involve several gigabytes of tract-level data and take several hours on the first run. Subsequent runs use a local cache and are much faster.
2. **Building the adjacency graph.** The algorithm converts the census tract boundary file into an adjacency graph, where tracts are nodes and shared boundaries are edges. This preprocessing step runs once per state-year combination and produces a binary adjacency file (`.adj.bin`) that is used for all subsequent algorithm runs.

3. **Validating contiguity.** Island tracts (tracts with no land-border adjacency to other tracts, such as coastal islands) require special handling. The **redist** system includes an island-detection routine that identifies such tracts and connects them to the nearest mainland tract for algorithmic purposes.

The total data preparation time for a single state is approximately 30 minutes to 4 hours depending on state size. For a 50-state run, total preparation time is several days but parallelizes efficiently across available computing resources.

6.2 Public Transparency

Democratic legitimacy for algorithmic redistricting requires that the public be able to understand what the algorithm does and why the output looks the way it does. This is a non-trivial communication challenge.

Several mechanisms promote transparency. First, the algorithm’s inputs (census population counts, tract boundary files) are public data. Any member of the public can download the same inputs and verify that the algorithm’s output matches what the adopting authority publishes.

Second, the algorithm’s criteria (population equality, compactness) are simple to explain and have been endorsed by courts as legitimate. A public-facing explanation need not expose the mathematical details of METIS graph partitioning; it can describe the algorithm as a map-drawing system that balances population across districts and minimizes the number of geographic boundaries crossed.

Third, the criteria-compliance report generated by **redist analyze** provides quantitative documentation of how well each criterion is satisfied. Publishing this report alongside the map gives the public verifiable evidence of criterion compliance.

The remaining challenge is explaining why the output looks the way it does in specific cases: why two cities are split, why a county is divided, why certain communities are in different districts. The honest answer is often that the algorithm drew the closest boundary consistent with population balance and compactness, and that the resulting configuration is not a deliberate choice but a geometric consequence. This explanation, while accurate, is unfamiliar and may be unsatisfying to communities that expected to be kept together.

6.3 The Challenge Process

Under all three pathways, parties can challenge the adopted map in court. The challenge process for an algorithmically produced map has several distinctive features.

No intent to challenge. Because the algorithm cannot access partisan data, challengers cannot allege partisan intent. They must rest their challenge on the output’s effects—whether the resulting districts dilute minority voting strength, fail to satisfy compactness requirements, or violate population equality. Effect-based challenges require more evidence and face a higher legal standard than intent-based challenges.

Reproducibility as a defense. Because the algorithm is deterministic given its inputs and parameters, the adopting authority can reproduce the output at any time and demonstrate that the same inputs produce the same result. This reproducibility is a strong defense against claims that the algorithm was manipulated: any alleged manipulation would be visible in the documented parameters.

Expert testimony on criteria compliance. Challengers will likely retain experts to argue that the algorithm failed to adequately satisfy one or more criteria. The adopting authority’s response is the criteria-compliance report plus the testimony of the technical expert who ran the algorithm. Unlike expert testimony about a legislator’s intent in drawing a traditional map, algorithmic expert testimony is grounded in reproducible quantitative output.

7 Conclusion

Algorithmic redistricting is implementable through three distinct pathways: commission adoption, statutory enactment, and court-ordered remedy. Each pathway places different institutional actors in the role of map adopter, but all three share a common structure: the algorithm produces a starting plan that satisfies quantifiable mathematical criteria, and human oversight bodies apply judgment to the residual questions (communities of interest, VRA specifics) that the algorithm cannot resolve without partisan data.

The Arizona case study illustrates that the commission pathway is legally established and operationally feasible. The AIRC’s constitutional authority is confirmed; its five-member bipartisan structure provides built-in balance; and its existing community-of-interest hearing process can absorb the algorithm output as a constrained starting plan without fundamental change to the commission’s role.

The California case study illustrates that the statute pathway is feasible where a state’s redistricting criteria map cleanly onto the algorithm’s weight parameters. California’s criteria—population equality, VRA compliance, county contiguity, compactness, communities of interest—correspond to documented features of the **redist** system. The primary implementation challenge is calibrating the county-sticky weight to California’s specific county structure, which varies enormously in size from Los Angeles County to Alpine County.

The North Carolina case study illustrates that the court-order pathway is the most legally stable of the three, because it places the burden of demonstrating constitutional defect on challengers rather than defenders. The special master structure is well-established in federal redistricting practice and available to state courts. The algorithm’s inability to access partisan data eliminates the most common basis for constitutional challenge—partisan intent—leaving challengers with the more demanding task of demonstrating constitutional violation through effects alone.

Common to all three pathways is the need for data preparation, public transparency, and a structured challenge process. These challenges are real but tractable. The **redist** system’s documented inputs, reproducible outputs, and quantitative criteria-compliance reports address transparency and challenge-process concerns directly. Data preparation, while time-consuming for large states, is a one-time cost that supports subsequent redistricting cycles.

Adoption of algorithmic redistricting at scale would likely follow the pattern of any significant institutional reform: early adopters in states with independent commissions or active state-court supervision, followed by broader uptake as the early cases establish legal precedent and operational best practices. The three pathways analyzed here provide the institutional templates for that process.

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